Echoes in Bronze: An Embodied XR Journey Through Ancient Sound, Craft, and Ritual

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Figure 1: The VR experience where participants explore the bronze bell's two-tone response in a virtual landscape.

Abstract

Echoes in Bronze is an archaeologically grounded virtual reality heritage experience created by archaeologists, curators, physicists, and computer scientists, through on-site 3D scans and sound capture at the artefact's site of discovery and museums. The work tells the story through a 1000 BCE bronze bell, following the artefact from its source of production, rituals, archaeological recovery, and interpretation. Participants carry the material for casting, smelt it, and pour the molten bronze into a mould, break the mould to reveal the instrument, and strike different acoustic regions in VR to experience its characteristic two-tone response. After VR, they can strike a scaled physical replica of the bell to experience different tones materially. Guided by the VR Dimensionality Framework (8Df), the work links bodily action, spatial sound, historical setting, and acoustic discovery. The project contributes an evidence-responsive XR workflow for converting archaeological documentation, acoustic measurement, landscape capture, and embodied interaction into multisensory cultural interpretation.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; • **Applied computing** → **Arts and humanities**; • **Information systems** → **Multimedia information systems**.

Keywords

Virtual Reality, XR, Digital Heritage, Virtual Archaeology



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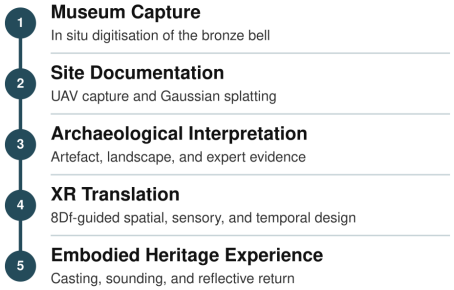


Figure 2: Evidence-to-experience workflow for embodied archaeological XR.

1 ARCHAEOLOGICAL EVIDENCE AS XR DESIGN MATERIAL

Our project demonstrates how archaeological evidence [Li et al. 2026] can actively inform the design of virtual environments and VR storytelling. Through casting, striking, and listening, participants gain an embodied understanding of the cultural significance and lived experience of pre-Qin bells. In collaboration with Tanheli Museum, we digitised the bronze bell, documented its acoustic properties, and captured the discovery landscape through drone imagery. The XR experience translates archaeological and scientific evidence into virtual time travel [Ch'ng 2009], situating the digitised bronze bell within reconstructed archaeological contexts that link museum artefact, discovery landscape, and embedded acoustic knowledge. These forms of evidence support a workflow from museum and site documentation through archaeological interpretation to embodied XR (Figure 2).

2 PARTICIPANT EXPERIENCE AND INTERACTION

The VR experience begins in a contemporary museum, where a bronze bell stands silent and fixed in time. Reaching out to touch the bell dissolves the museum into a reconstructed settlement around 1000 BCE, bringing the bell into a living world of craft, ritual, and belief. Participants use tracked VR controllers to carry the material for casting, smelt it, pour bronze into a mould, break the mould to reveal the bell, and enter a ritual landscape where they strike different acoustic regions of the bell. The interaction reveals the bell's two-tone response, allowing participants to understand ancient acoustic knowledge through action. A tracked VR controller drives a virtual mallet, and strike position triggers the corresponding measured tone in spatial audio. A brief controller vibration marks impact. The experience is guided by the VR Dimensionality Framework [Ch'ng 2026b] and human-in-the-loop co-ideation [Ch'ng 2026a], using presence as a foundation for cultural, cognitive, and emotional engagement. After VR, participants strike a scaled physical replica of the bell, extending the virtual discovery of two-tone response into direct tactile and acoustic experience.

Table 1: XR Demo Setup

Item	Details
Platform	Unity, XR Interaction Toolkit
Hardware	VIVE / Quest 3 with tracked controllers
Mode	One HMD participant; controller-driven virtual mallet; mirrored audience display
Duration	10–12 minutes
Space	Approx. 3 m × 3 m; scaled bell replica displayed alongside
Audio	Headphones or spatial speaker setup

3 EXTENDED REALITY VISION AND CONTRIBUTION

Our project contributes to Cultural Weaving and XR as Interwoven Mediality by connecting a museum-held artefact, an archaeological landscape, bronze-casting practice, ritual interpretation, embodied interaction, and spatial sound. Culture is presented not as a static visual archive, but as a situated relationship among object, place, action, sound, and memory [Cameron and Kenderdine 2007]. XR is essential because interpretation depends on the participant's position, movement, manual action, and acoustic discovery: the bell is virtually made, struck, and heard before being encountered again as a physical sounding object. The project integrates digitisation, fieldwork, consultation, acoustics, and VR design into an evidence-responsive heritage pipeline [Zhou et al. 2025]. Implemented in Unity with the XR Interaction Toolkit for VIVE/Quest 3, the work positions XR as a medium for embodied archaeological interpretation. The XR demo setup is summarised in Table 1.

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